

The commercial use of digital media to market alcohol products: a narrative review

Tim Lobstein¹, Jane Landon², Nicole Thornton³ & David Jernigan³

Public Health Advocacy Institute of Western Australia, Curtin University, Perth, Western Australia, Australia,¹ UK Health Forum, London, UK² and Johns Hopkins Bloomberg School of Public Health, Baltimore, USA³

ABSTRACT

Background and aims The rising use of digital media in the last decade, including social networking media and downloadable applications, has created new opportunities for marketing a wide range of goods and services, including alcohol products. This paper aims to review the evidence in order to answer a series of policy-relevant questions: does alcohol marketing through digital media influence drinking behaviour or increases consumption; what methods of promotional marketing are used, and to what extent; and what is the evidence of marketing code violations and especially of marketing to children? **Method and findings** A search of scientific, medical and social journals and authoritative grey literature identified 47 relevant papers (including 14 grey literature documents). The evidence indicated (i) that exposure to marketing through digital media was associated with higher levels of drinking behaviour; (ii) that the marketing activities make use of materials and approaches that are attractive to young people and encourage interactive engagement with branded messaging; and (iii) there is evidence that current alcohol marketing codes are being undermined by alcohol producers using digital media. **Conclusions** There is evidence to support public health interventions to restrict the commercial promotion of alcohol in digital media, especially measures to protect children and youth.

Keywords Advertising, alcohol, digital media, internet, marketing, youth.

Correspondence to: Tim Lobstein, Public Health Advocacy Institute of Western Australia, Curtin University, 10 Selby Street, Shenton Park, WA 6008, Australia. E-mail: tlobstein@iaso.org

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INTRODUCTION

Before the last decade, studies of the effects of alcohol advertising focused primarily on exposure to marketing through traditional media such as television, radio and magazines [1,2]. As potential consumers are using digital media increasingly for their everyday communications and entertainment [3], alcohol marketers have turned their attention to the opportunities these media provide. Brand owners and their agents have developed brand-promoting websites, placed brand-promoting pages on social media platforms such as Facebook, opened Twitter accounts and developed smartphone applications (apps), screensavers, video games, downloadable ringtones and events listings. They have increased their use of 'programmable' data-driven buying, which offers new efficiencies in reaching target audiences and works best on digital platforms [4]. Moreover, unlike traditional marketing, digital marketing provides marketers with opportunities to encourage continuing engagement and

individualized interaction with potential customers—for example, by hosting message boards and photo galleries where users can post comments or pictures—and can promote the re-distribution by potential customers of company-produced material, for instance through 'liking' or 'sharing' with their friends and through their own social networks.

Data from trade articles show that the Facebook platform alone can attract hundreds of thousands of 'likes' of popular brands and one brand alone, Heineken, had exceeded 20 million 'likes' globally by the end of 2015, while Budweiser, Smirnoff, Skol and Johnnie Walker all exceeded 12 million [5]. Two brands (Heineken and Chivas Regal) gained more than 500 000 new Facebook 'likes' in the month that included the Christmas–New Year partying period, 2015–16 [5].

Large numbers of people follow the Twitter accounts of alcohol companies, led currently by Smush Cocktails and Kirin beer at more than 500 000 followers each [5]. Company-run dedicated brand channels on YouTube

can also attract high numbers of viewings, with Budweiser's dedicated channel registering more than 160 million viewings globally, followed by Bud Light with more than 100 million viewings [5], although viewing by brand channel is the exception in how advertising is viewed on YouTube and the figures will underestimate the extent of exposure to YouTube-based branded messages [6].

Digital media offer opportunities for interactive engagement with media users. Social networking platforms (e.g. Facebook, Myspace, Bebo) and services that offer social messaging and image sharing (e.g. Twitter, Flickr, Tumblr, Instagram, WhatsApp) create opportunities for branded messages to be re-distributed rapidly and widely by the media users to their friends and social networks. This provides free publicity for the original message and, by being embedded in a user-endorsed medium, increases the likelihood that the message will attract a recipient's attention. Furthermore, digital brand promotion may be created entirely by media users without the brand owner's prior knowledge, with brand-promoting Facebook pages and YouTube videos uploaded without the brand owner's consent. While such material includes trademarked and copyrighted material, we are not aware of any actions taken by companies owning those trademarks and rights to have the material removed. Indeed, there is evidence that some companies encourage media users to create and upload home-made promotions featuring the companies' products [7].

Such informal, user-generated brand promotion is part of a broader discourse around alcohol use and drinking behaviour found widely on social networking sites, and contributing potentially to the construction of an individual's social identity and a group's social norms [8,9]. Griffiths & Casswell ([10], p. 525) suggested that 'young people are creating 'intoxigenic social identities' as well as 'intoxigenic digital spaces' that further contribute towards the normalisation of youth consumption of alcohol'.

While it may be the case that the alcohol producers encourage a continuing, positive 'buzz' around alcohol in social media, it is beyond the scope of the present paper to consider this wider discourse. Here we focus on digital communications that originate primarily from the brand owners and their agents, and which might be amenable to regulatory interventions. We focus on a set of questions relating specifically to the use of digital media by alcohol companies for the promotion of alcohol products, as follows:

- 1 Is there evidence that alcohol marketing through digital media influences drinking behaviour or increases consumption?
- 2 What methods of promotional marketing are used, and to what extent?

- 3 Is there evidence of marketing code violations and especially of exposure of children and adolescents to alcohol marketing promoted through digital media?

METHODS

Although the intention of this paper is to provide a narrative review, we approached the search for material as if it were a full systematic review, and included in our search both peer-reviewed and authoritative grey literature (i.e. fully referenced documents published by government agencies and government-funded organizations, independent non-governmental specialist institutions, professional associations and universities). Excluded from the search were books and chapters in books. The search terms used were alcohol + AND (advertis + OR marketing) AND [Internet OR (digital AND media) OR smartphone OR apps OR (social AND network+)]. The search was limited to English language literature, published January 2000 to December 2015. The year 2000 was taken as a practical historical cut-off, although we are aware of grey literature studies prior to this period, such as the reports by the Center for Media Education in the late 1990s which sought to identify the 'new threats' posed to youth by access to alcohol promotion through digital media [11–13].

Searches were conducted through MEDLINE (PubMed), Proquest, Web of Science, the Cochrane Database, Embase/Ovid and Google general index. Peer-reviewed and grey literature was supplemented by hand-searching recent papers and consulting with authors of other papers in the present Supplement.

RESULTS

The numbers of papers identified are shown in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) diagram (Fig. 1). Grey literature was identified primarily in the Embase and Google searches. For Embase, the top two levels of relevance were examined. In Google the search terms gave more than 30 000 potential records, and the first 500 of these (sorted by relevance) were examined by title and 20 items selected for further inspection.

Following elimination of duplicates, a total of 231 abstracts were inspected, 172 of which failed to meet the inclusion criteria. From the remaining papers, a further 38 titles were identified from hand-searching the references, providing a total of 97 documents subjected to detailed examination. Of these, 67 were peer-reviewed studies and 30 were grey literature documents. Inspection of the detailed text led to a further 50 documents being removed for failing to meet the inclusion criteria (due primarily to insufficient quantitative data or lack of specific

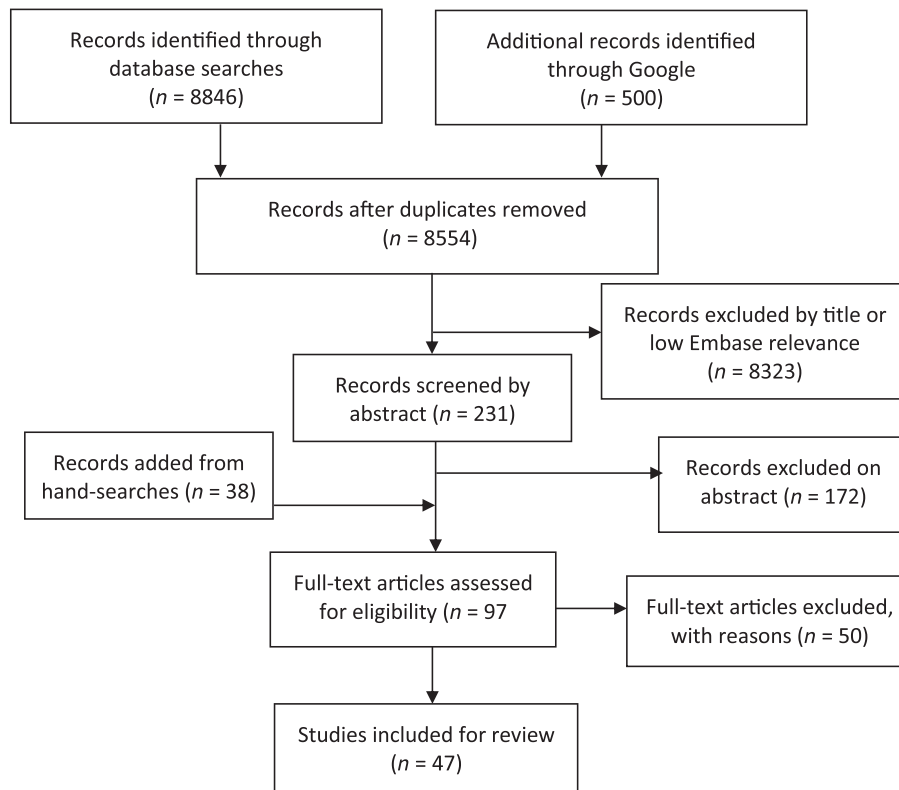


Figure 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of articles reviewed

information on commercial marketing), leaving 47 papers (33 peer reviewed and 14 grey literature) included in the current review.

FINDINGS

A short narrative description of each of the papers included in this review is provided in the online Supporting Information tables. Here we summarize the content in response to the questions posed in the introduction.

Is there evidence that alcohol marketing through digital media influences drinking behaviour or increases consumption?

Five documents were identified (see Supporting information, Table S1). All of them reported primary data, and in each case the evidence supported the hypothesis that exposure to alcohol marketing in digital media is associated with alcohol purchase intentions, drinking patterns or higher levels of alcohol consumption.

Four of the studies [14–17] used individual interviews to estimate recognition of digital marketing and self-reported alcohol consumption patterns. Three of the four studies were limited to youth aged less than 21 years, and of these, two were restricted to children aged less than 17 years. Drinking behaviour associated with exposure to

internet marketing included increased consumption, regular consumption, risky behaviour and binge drinking, but not initiation of drinking.

None of the four studies used longitudinal designs or controlled interventions, and therefore could not determine the direction of causality.

The fifth study [18] examined the effect of internet marketing in states in the USA where regular marketing was restricted compared with states where it was not, and found that exposure to internet marketing increased the proportion of residents (adults) in the restricted states who said they would purchase alcohol, indicating that digital marketing exposure undermines traditional marketing restrictions. The authors estimated that advertising on the internet reduced the effect of a ban on traditional advertising by 62%.

What methods of promotional marketing are used, and to what extent?

The literature search identified 22 studies (reported in 26 publications) that described the nature and magnitude of promotional alcohol marketing on digital media (see Supporting information, Table S2). All studies found extensive evidence of alcohol promotion by leading brands and found a variety of approaches being used to encourage engagement with a brand and the

re-distribution of both branded and general pro-consumption messages, including:

- interviews with celebrities
- free downloads of music, pictures and messaging accessories
- notices of parties, and competitions for tickets to enter party events
- notices of events (sports, music)
- free downloadable apps for entertainments, games, hangover advice, alcohol intake monitoring and local retailer location
- free downloadable widgets to aid invitations to parties via smartphones
- opportunities to upload messages and photos to the brand website or Facebook page
- opportunities to purchase branded merchandise and collectables
- opportunities to purchase alcohol for direct delivery
- competitions for user-generated videos, cocktail recipes, apps and games
- tacit approval for unofficial brand-promoting Facebook pages (some brands have more than 50 of these)
- social media being used by local alcohol outlets, bars and pubs to encourage clients' messaging and photo-sharing.

Uzunoglu [19] noted that a successful strategy for alcohol marketers uses digital media to exploit interactivity, both 'business-to-consumer' and 'consumer-to-consumer'. As Carah ([20], p. 5) comments: "These interactions make brands a part of everyday conversations on Facebook and allow [the brands] to become part of the continuous 'flow' of content in the news feeds of their followers. With each interaction, brand content appears in the news feeds of their followers' friends".

Examination of 70 user-uploaded YouTube videos depicting alcohol intoxication [21] noted in passing that brand references were relatively high (44%), but there was no evidence of industry influence and no explicit advertising messages. While the videos commonly associated the intoxication with humour, they also occasionally depicted negative clinical outcomes.

Is there evidence of marketing code violations and especially of exposure of children and adolescents to alcohol marketing promoted through digital media?

Brand owners state that their marketing is designed to target adults, and not children or adolescents. This section describes evidence that marketers are creating material that is attractive to younger age groups, and by design or by neglect allowing younger age groups to gain access to their marketing.

Nine studies were found which gave evidence of exposure to digital alcohol marketing in certain groups, from which the authors conclude that these groups are

likely to be specifically targeted or may be specifically vulnerable to marketing messages (see Supporting information, Table S3).

De Bruijn's study of children's digital media usage in four European countries [22] found that, among children aged 13–14 years, 33% had received promotional e-mails, or joke, chain or wind-up emails mentioning alcohol brands; 18% had downloaded a mobile phone or computer screensaver containing an alcohol brand name or logo; and 66% had noticed an internet page that contained an alcohol advertisement.

A study of marketing exposure in different age groups interviewed Brazilian children aged 14–17 years and young adults aged 18–25 years, in 2005–06 [23]. The study found that 61% of the younger sample and 60% of the older sample reported exposure to alcohol advertising in a range of media 'almost every day' or 'more than once a day' in the previous month, although fewer than 10% in both age groups reported seeing alcohol advertising on the internet.

A study in the United Kingdom estimated the exposure of children (6–14 years) and young people (15–24 years) to alcohol marketing on social media, based on artificially created subscriptions to social networking sites [24]. It found that Facebook pages could not be accessed by subscribers who had registered as being under 18 years, but YouTube and Twitter content could usually be accessed by users of all ages.

The literature search identified a further 12 papers that provide evidence of various forms of code violation (see Supporting information, Table S4). The transgressions depend upon the code in place in a given region or country: the paper by Brodmerkel & Carah [25] provides evidence that alcohol brands available in Australia own Facebook pages which encourage viewers to interact with the page and to make statements that the company itself cannot make under the prevailing Australian code of marketing. The pages also display users' photos that encourage excessive consumption and indicate sexual advantages through alcohol consumption (both prohibited under the Australian code). The photos appear to show people aged under 25 years consuming alcohol, which is also in breach of the code.

Two papers from the United States [26,27] show that alcohol retailing through digital media allowed underage purchases to be delivered directly to users. The majority of retailer sites required only a confirmation that the purchaser was of the required age. Several sites claimed and utilized delivery services that were supposed to examine age verification at the time of delivery, but the researchers found that 40% of actual deliveries failed to ask for such verification. Furthermore, some sites claimed to use the United States Postal Service (USPS) to verify age at delivery, but USPS did not offer such a service at

the time of the research study; furthermore, it was (and remains) illegal to ship alcohol via the USPS.

A study in the United Kingdom [28] found that age verification was required on the home web pages of all 14 brands examined, and 11 were abiding by Advertising Standards Authority regulations stating that reference to responsible drinking should be made on the home pages of official alcohol brand websites. The prominence of these references varied, with some websites providing the Drinkaware logo and additional information, while others depicted the Drinkaware web link in small text at the foot of the home page.

A study by Gordon [29] in Scotland looked at 40 leading brands and found they had 17 promotional websites, all of which had age restriction messages or controls. However, many had content that the author believed broke industry codes by appealing to minors (with arcade games, pinball and Ninja games, along with colour graphics and music downloads). There were examples of implied sexual success (guides to flirting, or for a girls' night out, and a game in which players can create the perfect man or woman). This study also found evidence linking the brand to driving cars, with Formula 1 coverage (Budweiser), the World Rally Championship (VK Vodka) and an interactive game in which players can 'Max' a motor by modifying it (WKD).

DISCUSSION

In order to support policymaking, this paper sought to answer three questions: do digital media marketing messages influence drinking behaviour, what methods are used and are youth being targeted and marketing codes being violated? Five papers provided evidence that branded alcohol promotion in digital media is associated with increased alcohol consumption, or increased likelihood of binge drinking behaviour, based on six cross-sectional studies and one controlled intervention. The cross-sectional studies cannot show the direction of causality (e.g. drinkers may be more interested in alcohol advertising) or fully eliminate other factors that could influence simultaneously both digital marketing engagement and alcohol consumption (e.g. personality type). However, the controlled intervention study [18] indicated causality, and the authors of the study concluded that digital media marketing acted to increase consumption directly, rather than as a complementary activity to marketing through other media. They found that digital media marketing was especially effective for products that were newly launched or little-recognized in a particular region.

Twenty-two studies, reported in 26 papers, provided evidence of the nature and methods of promotional marketing in digital media, indicating a wide range of approaches used to attract and engage digital media users,

through both commercially owned media and social networking media. An alcohol brand may have as many as 50 different Facebook pages, along with YouTube clips and dedicated YouTube channels, and multiple Twitter feeds which all encourage interactivity and promote a general discourse on drinking behaviours.

Nine studies gave specific but limited evidence of the extent to which identifiable population groups are attracted to digital alcohol marketing. No study gave information on differential access or engagement according to social class or household income, or educational level or other indicator of socio-economic status. One paper indicated that some digital marketing appears to be directed specifically to Latino and African Americans in the United States.

Lastly, 12 studies, reported in 16 papers, gave evidence for apparent violations of marketing codes in digital media, primarily showing underage access to advertising and marketing messages on digital media, and to retail deliveries purchased through website promotions. Examples are given of online brand websites using material attractive to minors, linking the brand to sexual success, to car driving and to using violence and antisocial behaviour.

Limitations of the present paper reflect some of the limitations of the studies reviewed. Studies showing cross-sectional evidence linking digital media exposure to alcohol consumption are unable to show causality. Adult researchers investigating the experiences of underage youth may not be permitted by their institution's ethical committee to misrepresent themselves as a child. Papers describing digital media practices are specific to the time of the survey and may be overtaken by changing fashions and brand preferences, and evolving digital technology, so that survey results in 2010, for instance, may be out of date by 2016. Codes of marketing have evolved during the period covered in this review, as have methods for age verification and restricting access to websites. Lastly, the number of studies available on which to base a conclusion was small, but the findings showed notable consistency despite the heterogeneity of research methods and the variety of national contexts and cultural norms.

In the majority of papers examined the authors concluded that there was a need for policies to control and restrict alcohol promotion, and especially to protect underage youth from commercial incentives to engage in drinking behaviour. Proposals included regulatory restrictions on access to websites, website content and requirements to report website usage, supplemented with stronger industry codes and tougher sanctions [20,25,30–35]. To date, it appears that policy responses have been weak. In response to complaints of code-breaking on brand websites, the Australian Advertising Standards Bureau determined in 2012 [32] that a brand's Facebook page is a marketing communication tool and that all content on the page falls

under the industry's self-regulatory code of ethics, including consumer-created content such as user-generated comments and photos. The Alcohol and Tobacco Tax and Trade Bureau in the United States took a similar view in their Industry Circular of May 2013 [36]. It remains to be seen how these decisions will affect the content of brand-owned sites and whether they might be extended to the use by consumers of trademarked or copyrighted material without the owner's consent.

A mechanism which has been advanced for restricting minors' access to adult content has been the use of parental control software. An investigation by the Center on Alcohol Marketing and Youth in 2004 [37] showed that most software products were not adequate for protecting children from accessing alcohol-related websites: 76% of alcohol brands eluded parental controls half the time or more. However, the survey found that some software controls were highly effective, indicating that it was feasible at that time to control access to certain online content through parental oversight.

Lastly, it may be helpful to comment upon research priorities for further policy development. Meier [38] has noted that research has tended to rely on simplified models of marketing and has focused disproportionately on youth populations. The effects of cumulative exposure across multiple marketing channels, targeting of messages at certain population groups and indirect effects of advertising on consumption remain unclear. Meier highlighted specific evidence needs for policymakers:

- the size of marketing effects for the whole population and for policy-relevant population subgroups
- the balance between immediate and long-term effects and the time lag, duration and cumulative build-up of effects
- the comparative effects of partial versus comprehensive marketing restrictions on consumption and harm.

As engagement with marketing grows [39], there is a need for longitudinal data identifying brand-specific youth exposure and brand-specific alcohol consumption in order to demonstrate more convincingly the causal links from marketing activity to drinking behaviour.

CONCLUSIONS

This paper summarizes the research literature available since 2000 on alcohol marketing in digital and social media. The literature indicated that marketing through digital media is likely to be having an impact upon drinking behaviour, that the marketing activities make use of materials and approaches that are attractive to young people and that current marketing codes are likely to be undermined by digital media.

The rapid development of immersive technologies that promote brands and encourage user engagement, and participation in the marketing itself, raises important and

challenging regulatory questions. The studies reported here show a consistency of findings to support the suggestion that digital marketing is effective and attractive to youth, and indicate a need for policymakers to respond to these questions.

Declaration of interests

None.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Table S1 Association between digital marketing and alcohol behaviour.

Table S2 Nature and extent of digital alcohol marketing.

Table S3 Targeting and exposure of demographic groups.

Table S4 Evidence of marketing code violations.

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